PVsyst V8.1.7

# PVsyst - Simulation report

## Grid-Connected System

Project: cuiaba

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 2400 Wp

Cuiabá - Brazil

**Author**

PVsyst TRIAL

**PVsyst V8.1.7**
VC0, Simulation date:
16/10/26 22:03
with V8.1.7

## Project summary

**Geographical Site**
**Cuiabá**
Brazil

**Situation**

| | |
|---|---|
| Latitude | -15.5987 °(S) |
| Longitude | -56.0991 °(W) |
| Altitude | 188 m |
| Time zone | UTC-4 |

**Project settings**
Albedo 0.20

**Weather data**
Cuiabá
NASA POWER satellite data 2001-2020 - Synthetic

## System summary

**Grid-Connected System**
**No 3D scene defined, no shadings**

**Orientation #1**
**Horizontal plane**

**Near Shadings**
no Shadings

**User's needs**
Unlimited load (grid)

**System information**

| PV Array | | Inverters | |
|---|---|---|---|
| Nb. of modules | 4 units | Nb. of units | 1 unit |
| Pnom total | 2400 Wp | Total power | 2 kWac |
| | | Pnom ratio | 1.20 |

## Results summary

| | | | | | |
|---|---|---|---|---|---|
| Simulation step | Hourly | | | | |
| Produced Energy | 3872.6 kWh/year | Specific production | 1614 kWh/kWp/year | Perf. Ratio PR | 83.98 % |

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## General parameters

**Grid-Connected System** — **No 3D scene defined, no shadings**

**Simulation step**
Hourly

**Orientation #1**
**Horizontal plane**

**Models used**

| | |
|---|---|
| Transposition | Perez |
| Diffuse | Perez, Meteonorm |
| Circumsolar | separate |

**Horizon**
Free Horizon

**Near Shadings**
no Shadings

**User's needs**
Unlimited load (grid)

## PV Array Characteristics

**PV module**

| | |
|---|---|
| Manufacturer | Generic |
| Model | JKM-600N-72HL4-V |
| (Original PVsyst database) | |
| Unit Nom. Power | 600 Wp |
| Number of PV modules | 4 units |
| Nominal (STC) | 2400 Wp |
| Modules | 1 unit x 4 In series |
| **At operating cond. (50°C)** | |
| Pmpp | 2226 Wp |
| U mpp | 163 V |
| I mpp | 14 A |

**Inverter**

| | |
|---|---|
| Manufacturer | Generic |
| Model | MIC 2000TL-X |
| (Original PVsyst database) | |
| Unit Nom. Power | 2.00 kWac |
| Number of inverters | 1 unit |
| Total power | 2.0 kWac |
| Operating voltage | 50-500 V |
| Pnom ratio (DC:AC) | 1.20 |

**Total PV power**

| | |
|---|---|
| Nominal (STC) | 2.40 kWp |
| Total | 4 modules |
| Module area | 10.3 m² |
| Cell area | 9.5 m² |

**Total inverter power**

| | |
|---|---|
| Total power | 2 kWac |
| Number of inverters | 1 unit |
| Pnom ratio | 1.20 |

## Array losses

**Thermal Loss factor**
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

**DC wiring losses**
Global array res. 196 mΩ
Loss Fraction 1.50 % at STC

**Module Quality Loss**
Loss Fraction -0.75 %

**Module mismatch losses**
Loss Fraction 2.00 % at MPP

**IAM loss factor**
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

| 0° | 30° | 50° | 60° | 70° | 75° | 80° | 85° | 90° |
|---|---|---|---|---|---|---|---|---|
| 1.000 | 0.999 | 0.987 | 0.963 | 0.892 | 0.814 | 0.679 | 0.438 | 0.000 |

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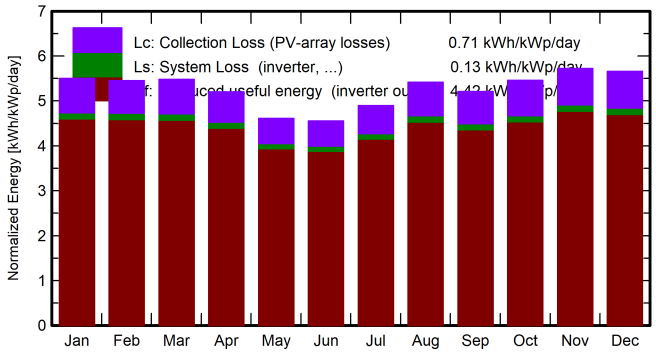
## Main results

### Simulation step
Hourly

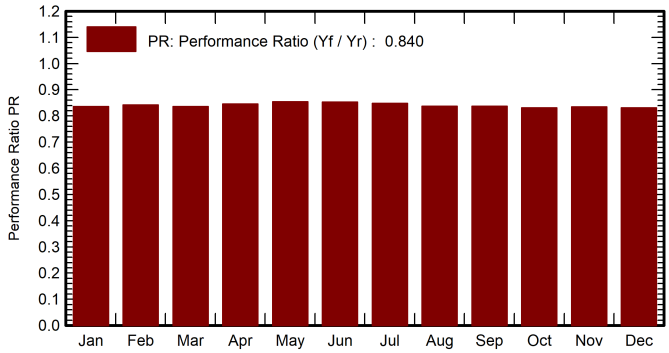
### System Production

| | | | |
|---|---|---|---|
| Produced Energy | 3872.6 kWh/year | Specific production | 1614 kWh/kWp/year |
| | | Perf. Ratio PR | 83.98 % |

**Normalized productions (per installed kWp)**




**Performance Ratio PR**




**Balances and main results**

| | GlobHor | DiffHor | T_Amb | GlobInc | GlobEff | EArray | E_Grid | PR |
|---|---|---|---|---|---|---|---|---|
| | kWh/m² | kWh/m² | °C | kWh/m² | kWh/m² | kWh | kWh | ratio |
| **January** | 170.7 | 81.25 | 26.65 | 170.6 | 166.3 | 352.9 | 342.3 | 0.836 |
| **February** | 152.9 | 69.99 | 26.41 | 152.7 | 148.9 | 318.0 | 308.7 | 0.842 |
| **March** | 169.9 | 73.64 | 26.45 | 169.9 | 165.7 | 351.1 | 340.7 | 0.836 |
| **April** | 156.1 | 57.41 | 25.76 | 156.0 | 152.0 | 326.2 | 316.5 | 0.845 |
| **May** | 143.1 | 46.46 | 24.09 | 143.0 | 138.5 | 301.8 | 293.0 | 0.854 |
| **June** | 136.8 | 36.64 | 24.03 | 136.7 | 132.1 | 288.1 | 279.7 | 0.852 |
| **July** | 151.8 | 35.72 | 24.58 | 151.8 | 147.1 | 318.4 | 308.9 | 0.848 |
| **August** | 167.9 | 46.05 | 27.25 | 167.9 | 163.5 | 347.8 | 337.5 | 0.837 |
| **September** | 156.4 | 61.94 | 29.54 | 156.3 | 152.7 | 323.7 | 314.1 | 0.837 |
| **October** | 169.3 | 74.26 | 29.47 | 169.3 | 165.5 | 348.1 | 337.7 | 0.831 |
| **November** | 171.7 | 75.59 | 28.10 | 171.7 | 167.8 | 354.2 | 343.7 | 0.834 |
| **December** | 175.4 | 82.41 | 27.10 | 175.4 | 171.2 | 360.5 | 349.7 | 0.831 |
| **Year** | 1922.2 | 741.35 | 26.62 | 1921.3 | 1871.3 | 3990.8 | 3872.6 | 0.840 |

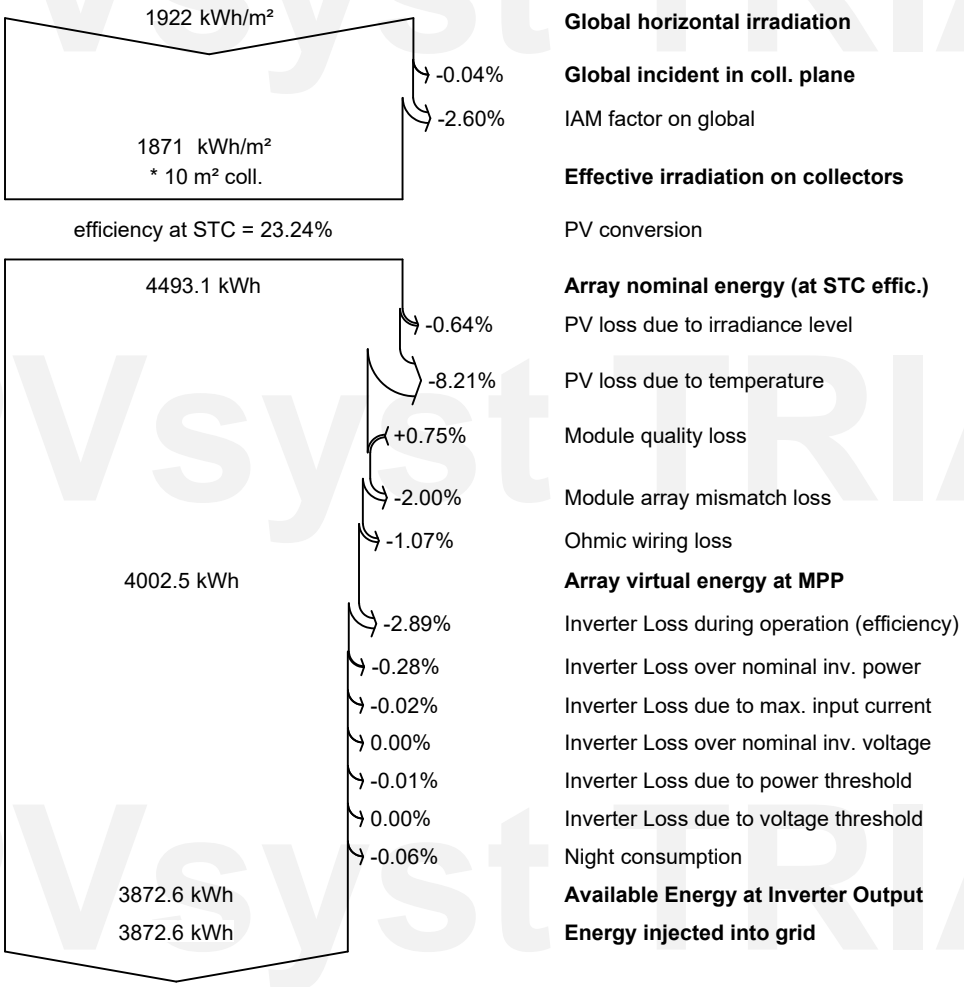
### Legends

| | | | |
|---|---|---|---|
| GlobHor | Global horizontal irradiation | EArray | Effective energy at the output of the array |
| DiffHor | Horizontal diffuse irradiation | E_Grid | Energy injected into grid |
| T_Amb | Ambient Temperature | PR | Performance Ratio |
| GlobInc | Global incident in coll. plane | | |
| GlobEff | Effective Global, corr. for IAM and shadings | | |

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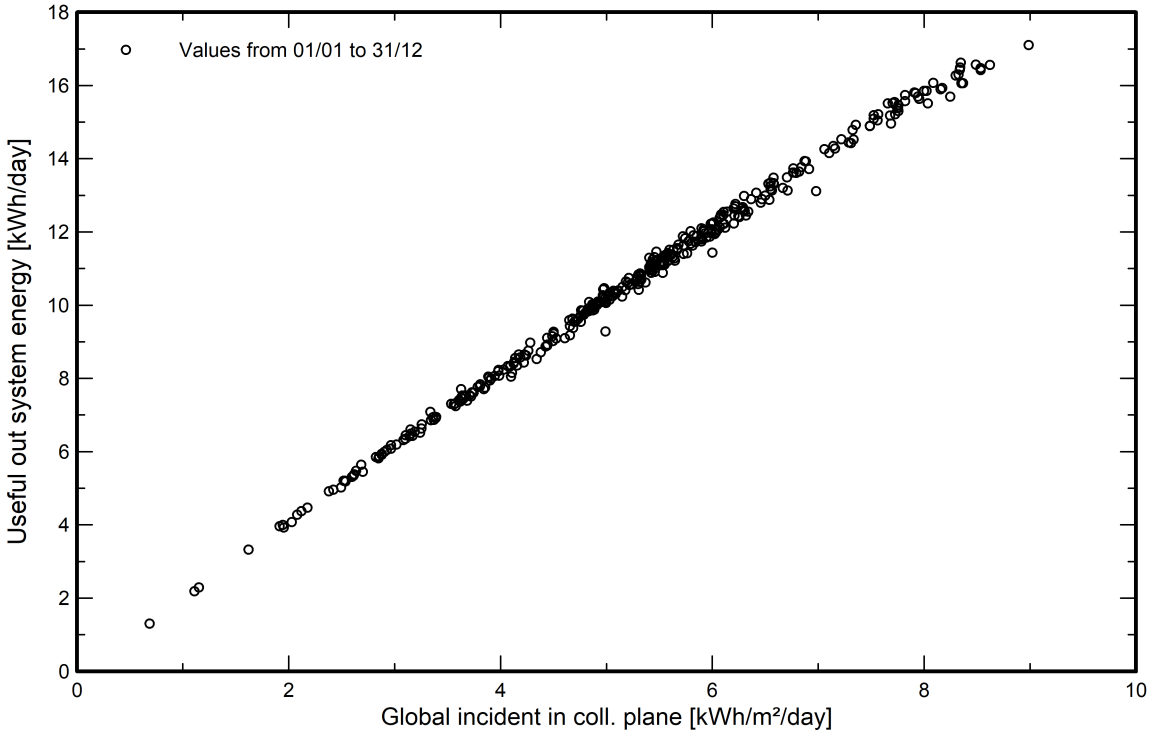
## Loss diagram



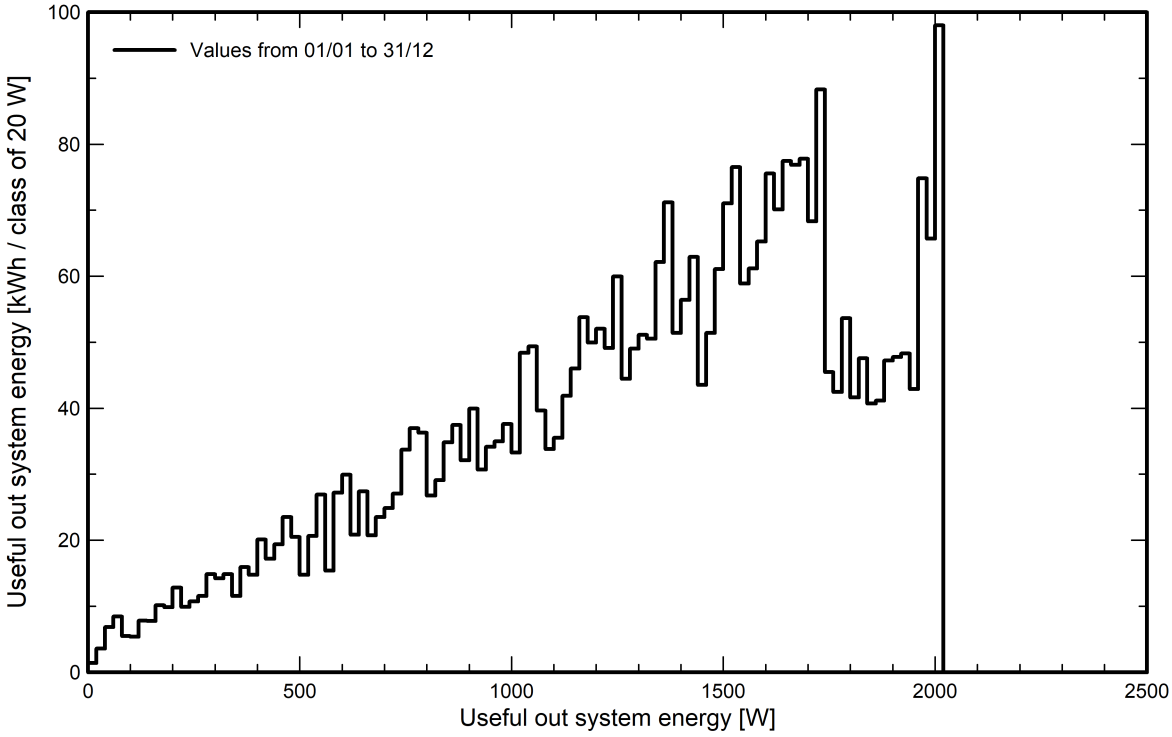
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## Predef. graphs

### Daily Input/Output diagram

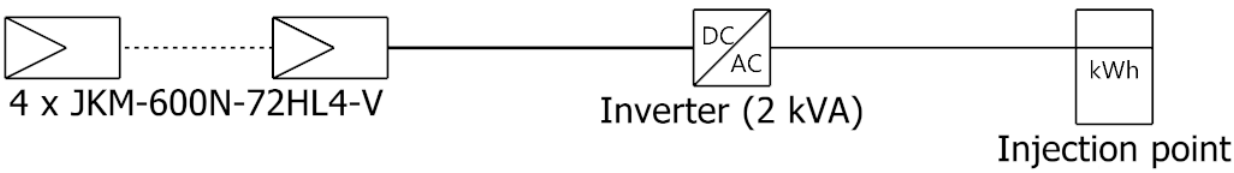



### System Output Power Distribution



# Single-line diagram

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| | |
|---|---|
| PV module | JKM-600N-72HL4-V |
| Inverter | MIC 2000TL-X |
| String | 4 x JKM-600N-72HL4-V |

| | | |
|---|---|---|
| | cuiaba | |
| | VC0 : New simulation variant | 16/10/26 |